RESEARCH STATEMENT

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INTRODUCTION

My research in geometric analysis concerns the relationship between the geometry of a Riemannian manifold and its global analytic properties. I am particularly interested in understanding which classical results of comparison geometry remain valid when their geometric hypotheses are weakened. Many foundational results rely on pointwise curvature bounds, which impose restrictions at every point of a manifold. Integral curvature conditions provide a different perspective by allowing curvature to deviate substantially from a prescribed bound in localized regions, but controlling the overall size of these deviations. It is natural to explore how much control this relatively weak geometric information has over the structure of a manifold.

My doctoral research at the University of Connecticut, under the supervision of Ovidiu Munteanu, investigates the spectrum of the Laplacian on complete, noncompact Riemannian manifolds. In particular, I study upper bounds for the bottom of the spectrum under integral bounds on the negative part of the Ricci curvature. My work establishes a Cheng-type spectral estimate under an integral curvature hypothesis, rather than the pointwise lower bound traditionally used to obtain it. The resulting paper, "Bottom Spectrum Estimate Under Curvature Integrability Condition," was published in the *Journal of Functional Analysis*.

More broadly, I am interested in the extent to which analytic information can translate into geometric control. The use of Green's functions in my doctoral work exemplifies this perspective as it provides a way to connect local geometric information to the global behavior of solutions of elliptic equations. I hope to continue exploring this interaction between geometry and analysis, both by identifying the limits of existing comparison results under weaker hypotheses and by developing techniques that make such results possible.

DOCTORAL RESEARCH: SPECTRAL ESTIMATES UNDER INTEGRAL CURVATURE BOUNDS

Background and motivation. The spectrum of the Laplacian encodes important information about the global geometry of a Riemannian manifold. In particular, the bottom of the spectrum, denoted by $\lambda_0(M)$, is the infimum of the spectrum of the nonnegative Laplace–Beltrami operator. It has variational characterization via a Rayleigh quotient:

$$\lambda_0(M) = \inf_{\varphi \in C_c^\infty(M)\setminus\{0\}} \frac{\int_M |\nabla \varphi|^2}{\int_M \varphi^2}.$$

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This quantity reflects the global behavior of the Laplacian and is closely connected to geometric properties such as volume growth, the existence of positive Green's functions, and the behavior of solutions to elliptic equations.

A fundamental result of Cheng [1] illustrates the relationship between curvature and spectral information. If M is a complete n-dimensional manifold whose Ricci curvature Ric satisfies

$$\mathrm{Ric} \geq -(n-1),$$

then

$$\lambda_0(M) \leq \frac{(n-1)^2}{4}.$$

Thus a pointwise lower bound on Ricci curvature imposes a universal upper bound on the bottom of the spectrum. The significance of this estimate is that a pointwise curvature condition constrains a global analytic quantity.

One question that can be asked is whether this conclusion persists when the curvature assumption is weakened. The classical lower bound rules out even isolated regions of curvature that fall below the prescribed threshold. In contrast, an integral condition can permit such regions, provided that the negative part of the curvature is sufficiently controlled in aggregate. This leads to a broader program of understanding which conclusions of comparison geometry survive when curvature is controlled in an averaged or integral sense.

There has been substantial work in this direction. Petersen and Sprouse [3] established comparison results under integral curvature assumptions, including results that recover Cheng's estimate in an appropriate limiting regime. Seto and Wei [4] subsequently obtained related estimates for the first eigenvalue of the p-Laplacian under integral curvature conditions. These results typically impose local integral controls with exponents tied to the dimension. My work approaches the problem with methods relying on properties of the Green's function, allowing us to investigate a regime in which the curvature control is not required to satisfy those same dimensional restrictions.

Main result. To measure the failure of the Ricci curvature lower bound in Cheng's theorem, define

$$w(x) = \left[\inf_{\substack{v \in T_x M \\ |v|=1}} \left\{ \mathrm{Ric}_M(v,v) + (n-1) \right\} \right]_- .$$

The function w measures the extent to which the Ricci curvature falls below $-(n-1)$, and the condition $w \in L^s(M)$ requires this curvature deficit to be integrable to the power s over the manifold.

The main theorem of my paper is the following.

Theorem 0.1. *Let (M^n, g) be a complete Riemannian manifold. If $w \in L^s(M)$ for some $s \geq \frac{3}{2}$, then*

$$\lambda_0(M) \leq \frac{(n-1)^2}{4}.$$

The conclusion is exactly the upper bound given by Cheng's theorem, but the hypothesis is different. Rather than requiring the Ricci curvature to stay above $-(n-1)$ at every point, we require only that the negative part of its deviation from this threshold belong to

$L^s(M)$ for $s \geq 3/2$. The result therefore demonstrates that a classical spectral comparison estimate can remain valid even when pointwise curvature control is lost.

The exponent in the integrability assumption is especially interesting. The condition $s \geq 3/2$ is independent of the dimension, unlike the dimensional thresholds that arise in several earlier local integral curvature results. This makes the theorem applicable in a different regime of geometric hypotheses. At the same time, the theorem does not claim that $s = 3/2$ is sharp; there may be room to improve this hypothesis further.

Ideas behind the proof. The proof uses the minimal positive Green's function for the Laplacian. The guiding idea is to assume that the desired spectral bound fails and then use the Green's function to derive a contradiction.

The first step is to obtain integral estimates for the Green's function and its gradient. These estimates, building on work of Li and Wang [2], describe the decay of the Green's function away from its pole when the bottom of the spectrum is positive. The second step is to use the Bochner formula, which relates the Laplacian of the gradient of a harmonic function to its second derivatives and the Ricci curvature. Applied to the Green's function, this identity brings the curvature deficit w directly into the analysis.

The central difficulty is controlling the resulting curvature term under an integral hypothesis rather than a pointwise bound. The assumption $w \in L^s(M)$, together with the Green's-function estimates and suitable integral inequalities, provides the control needed to carry out this argument. Combining these estimates with the variational characterization of $\lambda_0(M)$ yields the desired upper bound.

This approach illustrates a broader theme in geometric analysis: global analytic information can sometimes substitute for the local geometric control that classical comparison arguments require. Rather than estimating curvature pointwise, the proof uses the interaction between the Green's function, the spectrum, and integrability to control the effect of regions where the curvature lower bound fails.

Limitations and a complementary construction. The integrability threshold in the theorem is not known to be optimal. To explore the limitations of the result, I also construct examples of complete manifolds for which the same spectral conclusion fails under weaker integrability assumptions. More precisely, for every $s < 1/2$, the construction produces a manifold with $w \in L^s(M)$ but

$$\lambda_0(M) > \frac{(n-1)^2}{4}.$$

The examples are built using warped-product metrics, for which curvature can be expressed in terms of a single warping function. This reduces the geometric construction to a problem of arranging the curvature deficit and the global growth of the metric simultaneously.

These examples show that some integrability assumption is necessary, but they do not establish that the exponent $3/2$ in the main theorem is sharp. The gap between the sufficient condition $s \geq 3/2$ and the counterexamples for $s < 1/2$ leaves a range of exponents for which the answer remains unknown. This gap is among the most immediate directions for further work.

FUTURE DIRECTIONS

My first goal is to better understand the integrability threshold in my spectral estimate. **Theorem 0.2** establishes the desired upper bound under the assumption that the negative part of the Ricci curvature deficit belongs to $L^s(M)$ for $s \geq 3/2$, while the subsequent counterexample shows that the conclusion can fail for exponents below $1/2$. Can the sufficient integrability condition be improved, or can the counterexamples be refined to demonstrate failure under stronger curvature control? This would help separate the limitations of the current proof from genuine geometric obstructions.

A second direction is to investigate analogous questions for related elliptic operators, particularly the p-Laplacian. Existing spectral comparison results under integral curvature conditions provide a starting point for examining how the methods of my doctoral work might extend beyond the linear Laplacian. More broadly, I am interested in identifying which features of the Green's-function approach can be adapted to other operators or geometric hypotheses, and which require fundamentally different techniques.

Finally, I am interested in exploring questions motivated by Yau's conjecture on integral scalar curvature, particularly in light of recent counterexamples that have sharpened the question of which geometric hypotheses ensure such bounds. Work of Munteanu and Wang [5], which uses Green's-function comparison to establish scalar-curvature estimates in dimension three, provides a natural starting point for investigating related problems. I hope to explore whether these methods can be adapted to identify meaningful variants of such results.

REFERENCES

[1] S.-Y. Cheng, *Eigenvalue comparison theorems and its geometric applications*, Mathematische Zeitschrift 143 (1975), 289–297.

[2] P. Li and J. Wang, *Complete manifolds with positive spectrum*, Journal of Differential Geometry 58 (2001), 501–534.

[3] P. Petersen and C. Sprouse, *Integral curvature bounds, distance estimates and applications*, Journal of Differential Geometry 50 (1998), 269–298.

[4] S. Seto and G. Wei, *First eigenvalue of the p-Laplacian under integral curvature condition*, Nonlinear Analysis 163 (2017), 60–70.

[5] O. Munteanu and J. Wang, *Sharp Integral Bound of Scalar Curvature on 3-Manifolds*, Transactions of the American Mathematical Society, 2025.

[6] C. Durham, *Bottom spectrum estimate under curvature integrability condition*, Journal of Functional Analysis, 2025.